

ENTITY IDENTIFICATION

1. Customer

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- Customer_Name
- Gender
- Date_of_Birth
- Phone_Number
- Email
- Address
- Password

Relationships:

- Customer places Orders.
 - Customer gives Payments.
 - Customer uploads Prescriptions.
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2. Admin

Primary Key: Admin_ID

Attributes:

- Admin_ID (PK)
- Admin_Name
- Email
- Password
- Phone_Number

Relationships:

- Admin manages Medicines.
 - Admin manages Customers.
 - Admin manages Orders.
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3. Medicine

Primary Key: Medicine_ID

Attributes:

- Medicine_ID (PK)
- Medicine_Name
- Category
- Manufacturer
- Price
- Stock_Quantity
- Expiry_Date
- Batch_Number
- Description

Relationships:

- Medicine belongs to a Category.
 - Medicine appears in Order Details.
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4. Category

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name
- Description

Relationships:

- One Category contains many Medicines.
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5. Supplier

Primary Key: Supplier_ID

Attributes:

- Supplier_ID (PK)
- Supplier_Name
- Contact_Number
- Email

- Address

Relationships:

- Supplier supplies Medicines.
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6. Order

Primary Key: Order_ID

Foreign Keys:

- Customer_ID

Attributes:

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount
- Order_Status
- Delivery_Address

Relationships:

- Customer places many Orders.
 - Order contains many Medicines.
 - Order has one Payment.
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7. Order_Details

Primary Key: Order_Detail_ID

Foreign Keys:

- Order_ID
- Medicine_ID

Attributes:

- Order_Detail_ID (PK)
- Order_ID (FK)
- Medicine_ID (FK)
- Quantity
- Unit_Price

- Total_Price

Relationships:

- Connects Orders and Medicines.
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8. Payment

Primary Key: Payment_ID

Foreign Keys:

- Order_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Date
- Payment_Status
- Amount

Relationships:

- One Order has one Payment.
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9. Prescription

Primary Key: Prescription_ID

Foreign Keys:

- Customer_ID

Attributes:

- Prescription_ID (PK)
- Customer_ID (FK)
- Doctor_Name
- Hospital_Name
- Upload_Date
- Prescription_File

Relationships:

- Customer uploads many Prescriptions.

- Prescription may be linked to Orders containing prescription medicines.

10. Delivery

Primary Key: Delivery_ID

Foreign Keys:

- Order_ID

Attributes:

- Delivery_ID (PK)
- Order_ID (FK)
- Delivery_Date
- Delivery_Status
- Delivery_Address

Relationships:

- One Order has one Delivery.

Entity Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Order	One-to-Many (1:M)
Customer	Uploads	Prescription	One-to-Many (1:M)
Admin	Manages	Medicine	One-to-Many (1:M)
Admin	Manages	Order	One-to-Many (1:M)
Supplier	Supplies	Medicine	One-to-Many (1:M)
Category	Contains	Medicine	One-to-Many (1:M)
Order	Contains	Medicine	Many-to-Many (M:N) <i>(resolved through Order_Details)</i>
Order	Has	Payment	One-to-One (1:1)
Order	Has	Delivery	One-to-One (1:1)
Medicine	Appears In	Order_Details	One-to-Many (1:M)
Order	Contains	Order_Details	One-to-Many (1:M)

Primary Keys (PK)

- Customer_ID
 - Admin_ID
 - Medicine_ID
 - Category_ID
 - Supplier_ID
 - Order_ID
 - Order_Detail_ID
 - Payment_ID
 - Prescription_ID
 - Delivery_ID
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Foreign Keys (FK)

- Customer_ID → Order
 - Customer_ID → Prescription
 - Category_ID → Medicine
 - Supplier_ID → Medicine (*optional if one supplier per medicine*)
 - Order_ID → Order_Details
 - Medicine_ID → Order_Details
 - Order_ID → Payment
 - Order_ID → Delivery
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ER Diagram Flow

Customer

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Places

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Order ----- Payment

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Contains

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Order_Details

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Medicine ----- Category

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Supplier

Customer ----- Prescription

Order ----- Delivery

Admin ----- Manages ----- Medicine

Admin ----- Manages ----- Order

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